

Friedman Statistical Test Analysis for Segmentation Model Comparison

1. Objective

To determine whether the observed performance differences among the segmentation models are statistically significant, a Friedman non-parametric test was conducted using the Intersection over Union (IoU) scores obtained from five independent experimental runs.

The Friedman test is a rank-based statistical test designed for comparing multiple algorithms across repeated experiments. Unlike ANOVA, it does not assume normality of the data and is therefore widely used in machine learning and medical image segmentation studies.

2. Models Compared

The following segmentation models were included in the statistical analysis:

1. DenseUNet
2. DeepLabV3
3. PSPNet
4. SegFormer
5. UNet++
6. Proposed M-Net

The IoU values from five independent runs were used.

3. IoU Scores Used for Analysis

Run	DenseUNet	DeepLabV3	PSPNet	SegFormer	UNet++	M-Net
1	0.703144	0.675383	0.694825	0.593479	0.707716	0.780664
2	0.704717	0.672376	0.691442	0.569810	0.702213	0.783071
3	0.707213	0.663231	0.697808	0.573020	0.695091	0.774565
4	0.706830	0.664619	0.684569	0.609153	0.694940	0.778095
5	0.711487	0.672038	0.698383	0.606422	0.686490	0.782815

4. Rank Assignment

For each run, models were ranked according to their IoU score.

Rank 1 was assigned to the model with the highest IoU.

Rank 6 was assigned to the model with the lowest IoU.

Run 1 Ranking

Model	IoU	Rank
M-Net	0.780664	1
UNet++	0.707716	2
DenseUNet	0.703144	3
PSPNet	0.694825	4
DeepLabV3	0.675383	5
SegFormer	0.593479	6

Run 2 Ranking

Model	IoU	Rank
M-Net	0.783071	1
DenseUNet	0.704717	2
UNet++	0.702213	3
PSPNet	0.691442	4
DeepLabV3	0.672376	5
SegFormer	0.569810	6

Run 3 Ranking

Model	IoU	Rank
M-Net	0.774565	1
DenseUNet	0.707213	2
PSPNet	0.697808	3
UNet++	0.695091	4

Model	IoU	Rank
DeepLabV3	0.663231	5
SegFormer	0.573020	6

Run 4 Ranking

Model	IoU	Rank
M-Net	0.778095	1
DenseUNet	0.706830	2
UNet++	0.694940	3
PSPNet	0.684569	4
DeepLabV3	0.664619	5
SegFormer	0.609153	6

Run 5 Ranking

Model	IoU	Rank
M-Net	0.782815	1
DenseUNet	0.711487	2
PSPNet	0.698383	3
UNet++	0.686490	4
DeepLabV3	0.672038	5
SegFormer	0.606422	6

5. Average Ranks

The average rank for each model was computed as:

Average Rank = (Sum of Ranks) / Number of Runs

Model	Rank Sum	Average Rank
M-Net	5	1.0
DenseUNet	11	2.2

Model	Rank Sum	Average Rank
UNet++	16	3.2
PSPNet	18	3.6
DeepLabV3	25	5.0
SegFormer	30	6.0

6. Friedman Test Statistic

The Friedman test statistic is computed using:

$$\chi^2_F = (12N / k(k+1)) \times \Sigma(R_j^2) - 3N(k+1)$$

where:

N = 5 (number of runs)

k = 6 (number of models)

R_j = average rank of model j

Substituting the average ranks:

$$\Sigma(R_j^2)$$

$$= (1.0^2 + 2.2^2 + 3.2^2 + 3.6^2 + 5.0^2 + 6.0^2)$$

$$= 1 + 4.84 + 10.24 + 12.96 + 25 + 36$$

$$= 90.04$$

Therefore:

$$\chi^2_F$$

$$= (12 \times 5) / (6 \times 7) \times 90.04 - 3 \times 5 \times 7$$

$$= 23.63$$

7. Hypothesis Testing

Null Hypothesis (H0):

All models have equivalent performance.

Alternative Hypothesis (H1):

At least one model performs significantly differently.

The calculated Friedman statistic:

$$\chi^2 = 23.63$$

$$p\text{-value} = 0.00026$$

Significance level:

$$\alpha = 0.05$$

Since:

$$p\text{-value} < \alpha$$

$$(0.00026 < 0.05)$$

the null hypothesis is rejected.

8. Interpretation

The Friedman test revealed statistically significant differences among the evaluated segmentation models. The proposed M-Net consistently achieved the best rank across all five runs and obtained the lowest average rank (1.0), indicating superior and stable segmentation performance.

DenseUNet achieved the second-best average rank (2.2), followed by UNet++ (3.2) and PSPNet (3.6). DeepLabV3 and SegFormer obtained comparatively lower rankings.

These results demonstrate that the proposed M-Net significantly outperforms the competing segmentation architectures in terms of IoU performance across repeated experimental runs.

9. Conclusion

The Friedman statistical analysis confirmed that the observed improvements of the proposed M-Net are not due to random variation. With a Friedman statistic of $\chi^2 = 23.63$ and a p-value of 0.00026, M-Net achieved the highest and most consistent IoU performance among all compared models, establishing its superiority over the baseline architectures.